



Chemical Theory II

Statistical

Thermodynamics

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The course is developed based on the
Cambridge chemistry courses MELT

Summary of last lecture

- Nuclear spin statistics
 - Symmetry factor σ in $q_{\text{rot}} = kT / \sigma B$
 - Half the J states are missing in homonuclear diatomics.
 - In general, σ = number of superimposable rotated structures.
 - Some combinations of J and nuclear spin function are not allowed.
- Equilibrium constants from first principles
 - Equilibrium condition in terms of the chemical potential.
 - Chemical potential in terms of the partition function via Helmholtz energy.
 - Measure q_{elec} and q_{vib} from the bottom of the electronic ground state.
 - Measure the ground state relative to dissociated atoms.
 - K_c in terms of the volume-independent partition function, $f = q / V$.
 - K_p in terms of the partition function in the standard state, q° .

Adsorption of a gas on a surface

- Our last example is a little different as it deals with the equilibrium between the molecules in a gas and the same molecules adsorbed on a surface. The adsorption onto surfaces is commonly observed for many molecules and different types of surface. Adsorption is the first step in the heterogeneous catalysis of chemical reactions, a process of enormous commercial importance in the chemical industry.

Adsorption of a gas on a surface

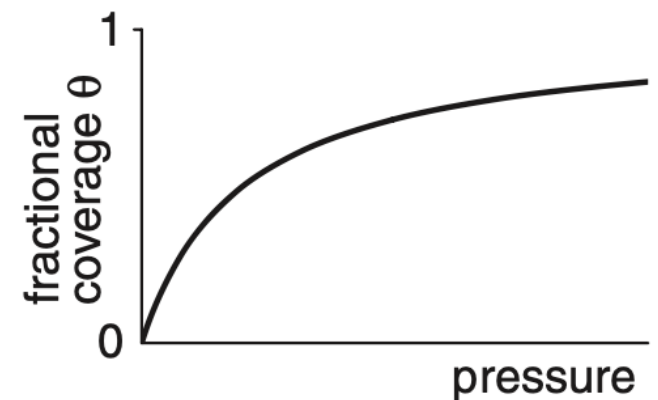
- A very simple model for this adsorption process is to assume that the surface has a number of equivalent sites each of which can be occupied by an adsorbed molecule. It is further assumed that the probability of a molecule being adsorbed at a particular site is independent of whether or not adjacent sites are occupied, and that only one molecule can be adsorbed at each site. Under these assumptions it is easy to use a simple kinetic argument to derive *the Langmuir isotherm which relates the fraction of occupied sites θ to the pressure p and a constant K_{ads} :*

$$\theta = \frac{K_{ads}p}{1 + K_{ads}p}$$

Adsorption of a gas on a surface

- *At low pressures, meaning $K_{\text{ads}}p \ll 1$, the term on the bottom of the fraction can be approximated to 1, so $\theta = K_{\text{ads}}p$ i.e. the fractional coverage increases linearly with pressure.*
- *At high pressures, meaning $K_{\text{ads}}p \gg 1$, the 1 on the bottom of the fraction can be ignored and so $\theta = 1$ i.e. all the sites are occupied.*
- The graph opposite shows how the initial linear rise in θ levels off to give complete coverage ($\theta = 1$) at sufficiently high pressures. It is possible to measure θ , for example by weighing the sample in *vacuo* and then in the presence of the gas; such an experiment features in the practical course.

$$\theta = \frac{K_{\text{ads}}p}{1 + K_{\text{ads}}p}$$



Derivation of the Langmuir isotherm

- We can derive the Langmuir isotherm by *first recognising that at equilibrium the chemical potential of the adsorbed molecules must be equal to that of the molecules in the gas phase*

$$\mu(\text{ads}) = \mu(\text{g})$$

- This derives from the same argument as for a chemical equilibrium $A \rightleftharpoons B$ for which the equilibrium condition is $\mu_A = \mu_B$.
- For the gas, the chemical potential can be written as

$$\mu(\text{g}) = \mu^\circ + RT \ln \frac{p}{p^\circ}$$

where μ° is the standard chemical potential i.e. the chemical potential when the gas is pure and at the standard pressure p° . This is a result from classical thermodynamics, as outlined previously.

Derivation of the Langmuir isotherm

- For the adsorbed molecules we are going to use the methods of statistical thermodynamics to first derive an expression for the Helmholtz energy **A** and then the chemical potential. Going right back to the start,

$$A = -kT \ln Q_N$$

gives the relationship between **A** and **Q_N**, the number of microstates of *the system (be careful to distinguish this from the partition function of a molecule)*

- The molecules absorbed onto the surface form our system. If we imagine that there are *M* surface sites, and that *N* molecules are absorbed onto this surface, then there are

$$\frac{M!}{N! (M - N)!}$$

ways of arranging these molecules on the surface.

Derivation of the Langmuir isotherm

$$\frac{M!}{N! (M - N)!}$$

- *It is important to note that in deriving this result we have assumed that the surface sites are distinguishable on the grounds* that they correspond to different physical positions on the surface, but that the molecules are indistinguishable.
- We also have to take account the energy levels available to the adsorbed molecule. If the molecular partition function of such a molecule is q , then the contribution of N such molecules to the number of microstates is q^N , as was discussed in earlier lectures. *Thus, the overall number of microstates is*

$$Q_N = q^N \times \frac{M!}{N! (M - N)!}$$

Derivation of the Langmuir isotherm

$$Q_N = q^N \times \frac{M!}{N! (M - N)!}$$

- Substituting this into $A = -kT \ln Q_N$ and using Stirling's approximation for the factorials gives

$$A = kT [-N \ln q - M \ln M + N \ln N + (M - N) \ln(M - N)]$$

- We can now compute the chemical potential from its definition

$$\mu = \left(\frac{\partial A}{\partial N} \right)_{V,T}$$

- Noting carefully that M and q are not functions of N , differentiation of the above expression for A with respect to N gives

$$\mu(\text{ads}) = kT \left[-\ln q + \ln \frac{N}{M - N} \right]$$

- The fraction of occupied sites, θ , is N/M , so this expression becomes

$$\mu(\text{ads}) = kT \left[-\ln q + \ln \frac{\theta}{1 - \theta} \right]$$

Derivation of the Langmuir isotherm

- At equilibrium we can equate $\mu(\text{ads})$ with $\mu(\text{g})$:

$$\begin{aligned}\mu(\text{g}) &= \mu(\text{ads}) \\ \mu^\circ + kT \ln \frac{p}{p^\circ} &= kT \left[-\ln q + \ln \frac{\theta}{1-\theta} \right]\end{aligned}$$

In writing this, R in the expression for $\mu(\text{g})$ has been replaced by k as we are working in molecular, rather than molar, quantities.

- Some careful rearrangement gives

$$b \frac{p}{p^\circ} = \frac{\theta}{1-\theta} \quad \text{where} \quad b = q \exp\left(\frac{\mu^\circ}{kT}\right)$$

- This can be rearranged to give an expression for θ

$$\theta = \frac{b(p/p^\circ)}{1 + b(p/p^\circ)}$$

Derivation of the Langmuir isotherm

$$\theta = \frac{K_{\text{ads}} p}{1 + K_{\text{ads}} p}$$

- This final expression is of the form of the Langmuir isotherm, with

$$K_{\text{ads}} = (q/p^\circ) \exp\left(\frac{\mu^\circ}{kT}\right)$$

- *The equilibrium constant which determines the surface coverage at a given pressure depends on the standard chemical potential of the gas, and the partition function of the adsorbed molecule.* The former could be expressed in terms of the partition function of the gaseous molecule, if we so wished.
- This is a very simple example of calculating $\mathbf{Q}_N$ directly, rather than working from the molecular partition function.

Course outline

- Rate constants

Rate constants

- Given the success of statistical thermodynamics in calculating equilibrium constants, the next logical step is to see if the theory can be applied to the prediction of rate constants.
- We will see in this section that by introducing the concepts of the potential energy surface and the transition state it is possible to devise a method of calculating rate constants using statistical thermodynamics. *Unfortunately the calculations are far from easy, so the theory is more used as a conceptual aid to understanding the factors which affect rate constants rather than as a practical tool for predicting them.*

Potential energy surfaces

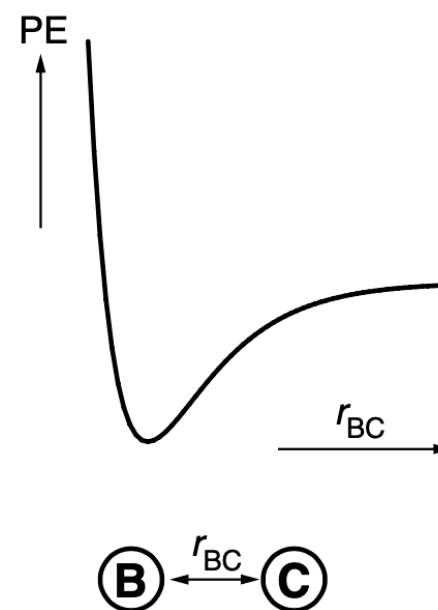
- Using molecular orbital (MO) theory we can calculate the electronic energy of any molecule we care to think of: *all we have to do is to specify the types and positions of the atoms.*
- When two molecules react all that really happens is that the electrons and nuclei rearrange themselves into new molecules. *During the reaction there are partially made bonds, but if we treat all of the nuclei and electrons in the reacting molecules together this is of no concern to our MO calculation.* We can therefore calculate the electronic energy of any arrangement, from the separated reactants right through the course of the reaction to the separated products.

Potential energy surfaces

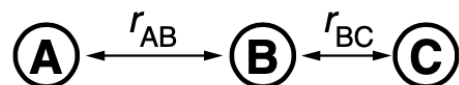
- To specify the positions of all of the nuclei throughout the reaction will require, even for relatively small molecules, rather a large number of co-ordinates. On top of this we probably need to consider many different directions and orientations with which the reactants can approach one another. The result of all these calculations is a *potential energy surface (PE surface or PES)*.
- *The PES gives us the energy of the arrangement of nuclei as a function of the very large number of co-ordinates needed to specify their positions.* It is rather hard to visualize or plot such a multi-dimensional function, but we can discuss its general features.

Potential energy surfaces

- *At certain positions on the PES there will be minima; these correspond to arrangements of atoms which give stable molecules.* They are stable in the sense that small distortions from these positions lead to an increase in the PE and hence a force which restores the molecule to its original position.
- *There will be minima corresponding to the reactants and the products.* We can imagine the reaction taking place by a point 'moving' from the position corresponding to the reactants to that corresponding to the products. We shall see that the route usually taken is rather a special one.



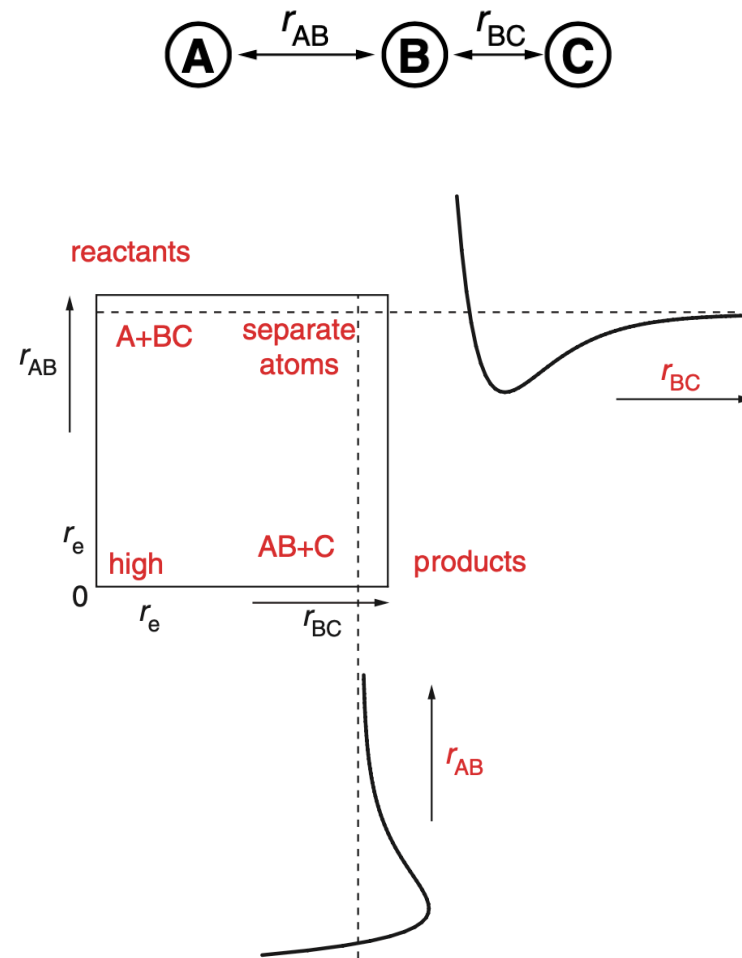
PES for the $A + BC \rightarrow AB + C$ reaction



- We will start out by considering the simplest possible reaction, that of an atom, A, reacting with a diatomic, BC. To simplify things even further we will restrict the motion of A to be along the line of B–C i.e. all three atoms are always in a line. Then only two distances, r_{AB} and r_{BC} are needed to specify completely the arrangement of the atoms and so the PE surface is just a function of two variables, and we stand a chance of being able to visualize it.

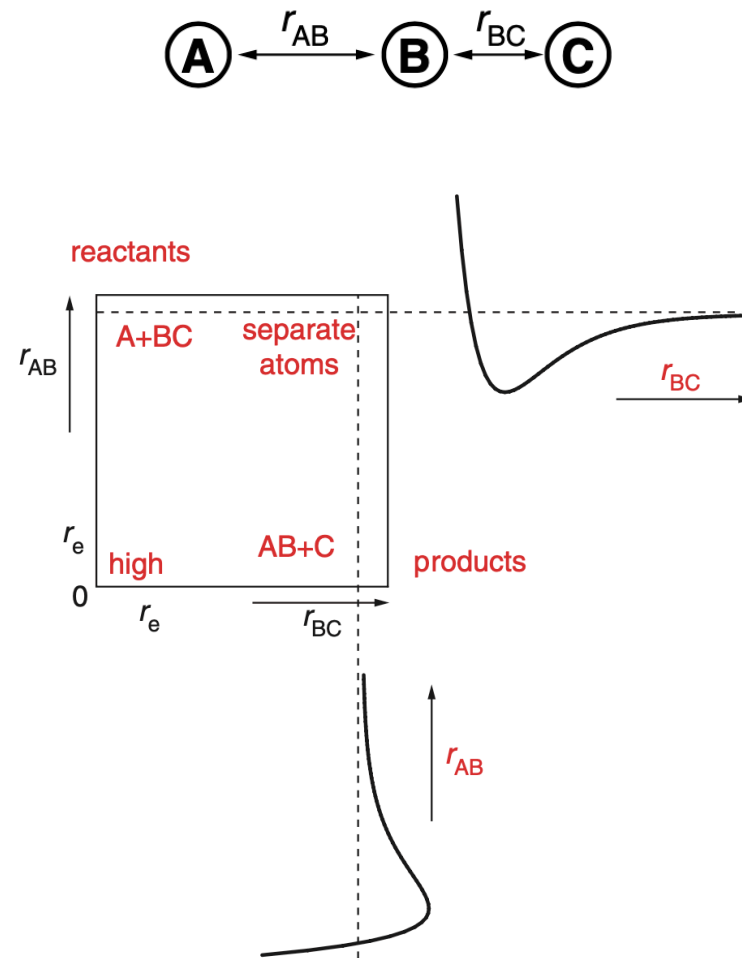
PES for the $A + BC \rightarrow AB + C$ reaction

- When A is a long way from BC – that is before the reaction takes place – there is no interaction between A and BC and so *the PE only depends on the distance r_{BC}* . As BC is a diatomic, the PE as a function of r_{BC} has the familiar form.
- When C is a long way from AB – that is after the reaction is over – there is no interaction between C and AB and so the PE only depends on the distance r_{AB} . The PE will vary as a function of r_{AB} in a similar way to that for BC.
- When A, B and C are all a long way away from one another, the PE is just the sum of the energies of the separate atoms. When A, B and C all get very close to one another we expect that the PE will rise steeply as the nuclei and core electrons begin to repel.



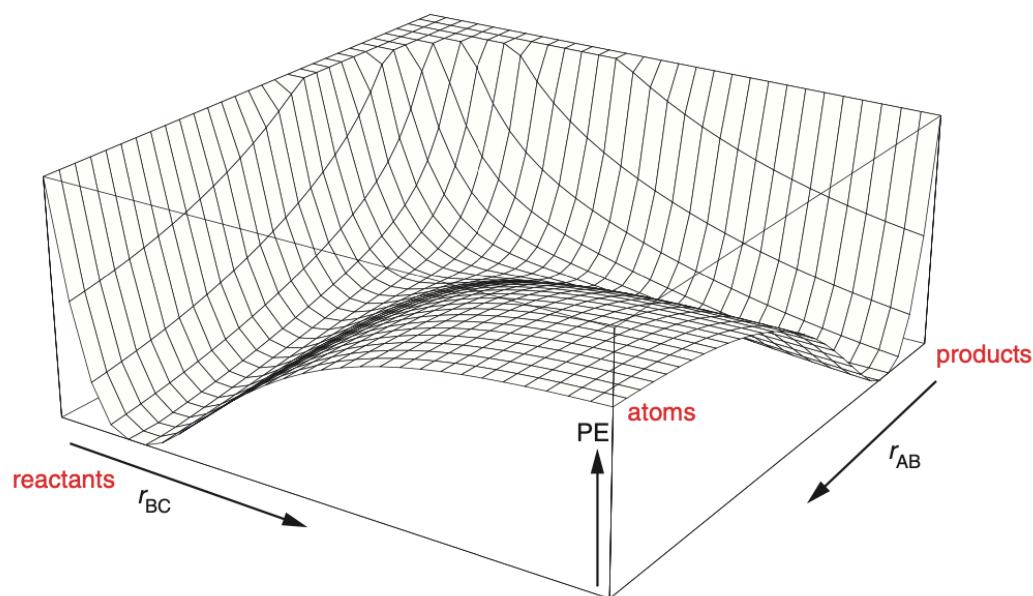
PES for the $A + BC \rightarrow AB + C$ reaction

- The surface is viewed from above, and in due course we will fill it in with contours. At this stage, however, we can only identify some general regions. When the atoms are pushed together, the potential rises steeply – marked ‘high’ on the diagram. When the atoms are all far apart, the PE is just that of the separate atoms – marked ‘separate atoms’.
- The reactants, $A + BC$, are found in the upper left part of the surface (r_{AB} large, $r_{BC} \approx$ equilibrium separation). The products, $AB + C$, are found in the lower right part of the diagram (r_{BC} large, $r_{AB} \approx$ equilibrium separation).
- The horizontal dotted line in the diagram indicates a cross-section taken at large r_{AB} . Likewise, the vertical dashed line indicates a cross-section taken when r_{BC} is large.

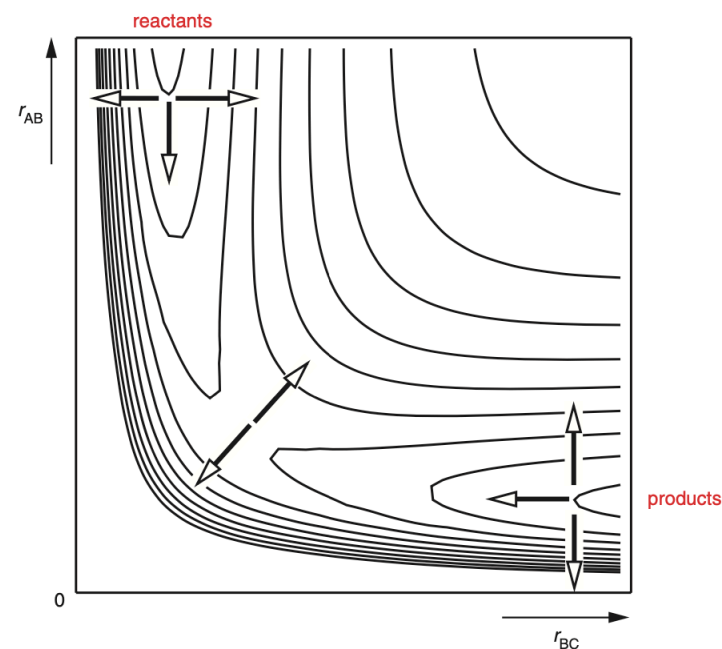


PES for the $A + BC \rightarrow AB + C$ reaction

- To fill in the rest of the details requires a complex calculation. A typical result is shown below, both in the form of a bird's eye view and a contour plot in which the contours correspond to points of equal potential.



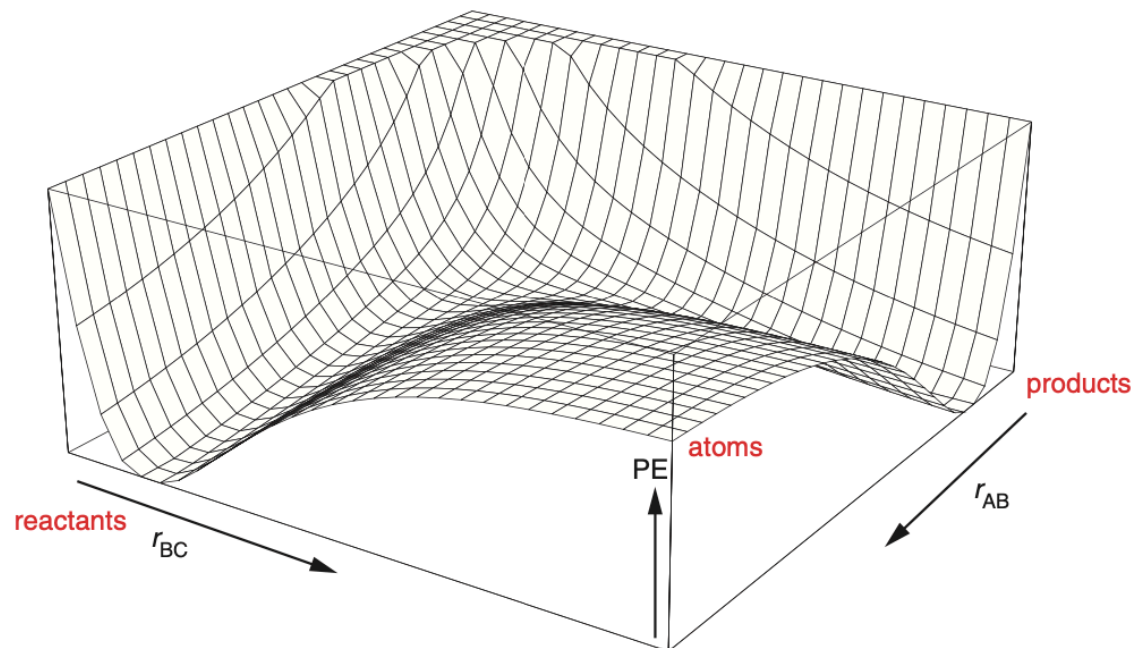
Perspective view of the PE surface for the $A + BC \rightarrow AB + C$ reaction. The arrangement consisting of the separated atoms is the corner coming towards us, while that corresponding to the atoms being compressed together is at the rear. Note how the cross-sections at large r_{AB} and r_{BC} are identical to the usual PE curves for diatomics.



Contour plot of the PES. The arrows point 'uphill' i.e. to regions of higher potential energy.

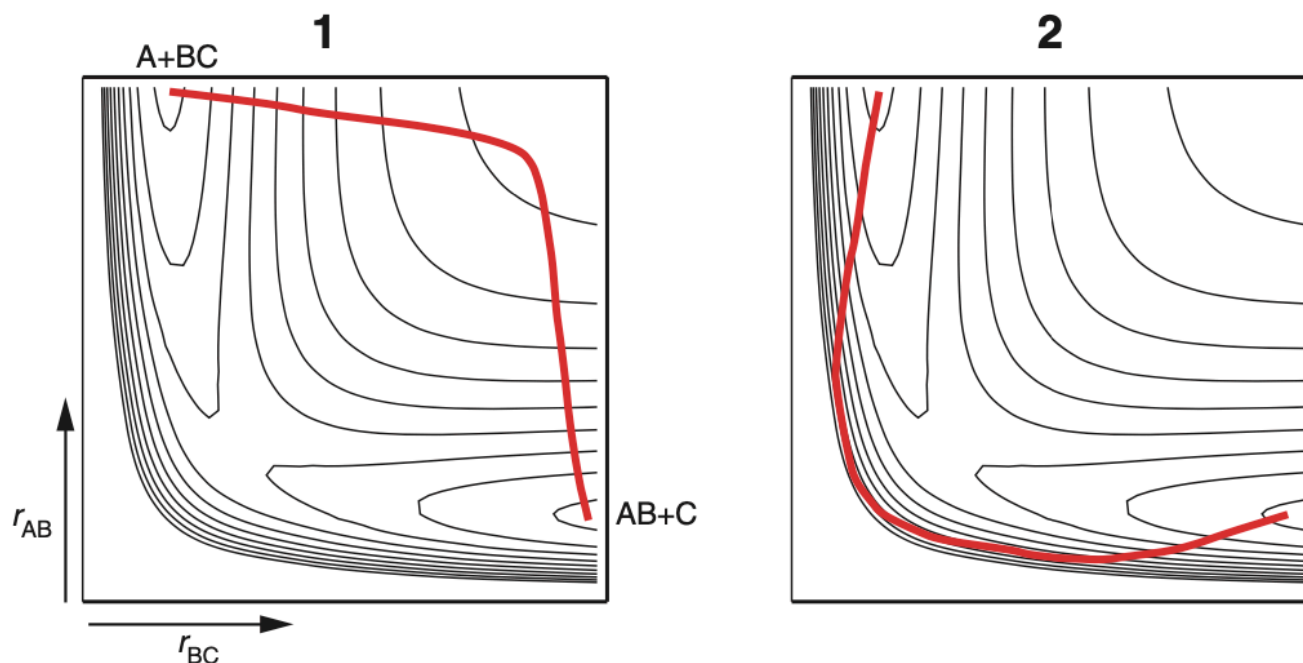
PES for the $A + BC \rightarrow AB + C$ reaction

- The form of the surface is two valleys meeting at right angles, one starting at the products and one at the reactants.* As we 'walk' up the valley away from the reactants the valley floor rises up; the same thing happens as we 'walk' away from the products. Somewhere in the middle the two valleys connect, forming a pass. If we started from the reactants we could walk up a valley, gaining height until we reached the head of the valley, then we would go over the pass and drop down into the other valley heading down towards the products.



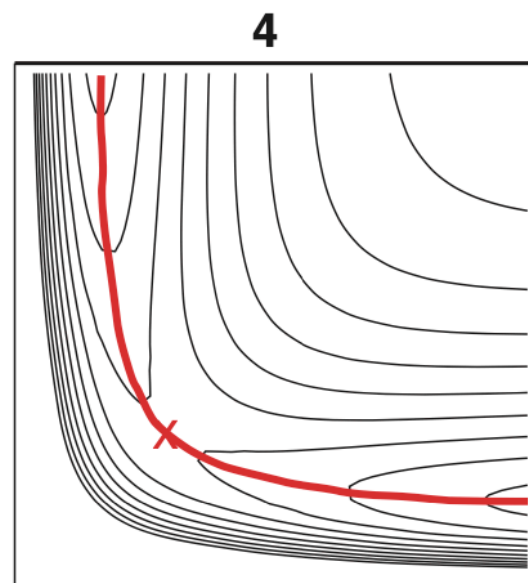
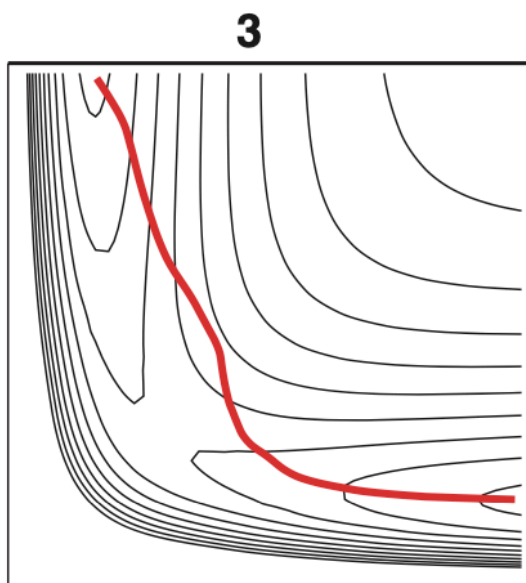
Pathways on the PE surface; the transition state

- There are very many ways of going from reactants to products on the PE surface.
- Route 1 (shown by the heavy line on the left-hand contour plot) involves first separating the BC molecule almost into atoms and then recombining A with B by bringing them closer together.
- Route 2 (shown in the right) involves pushing all three atoms together, and then letting them separate as products.



Pathways on the PE surface; the transition state

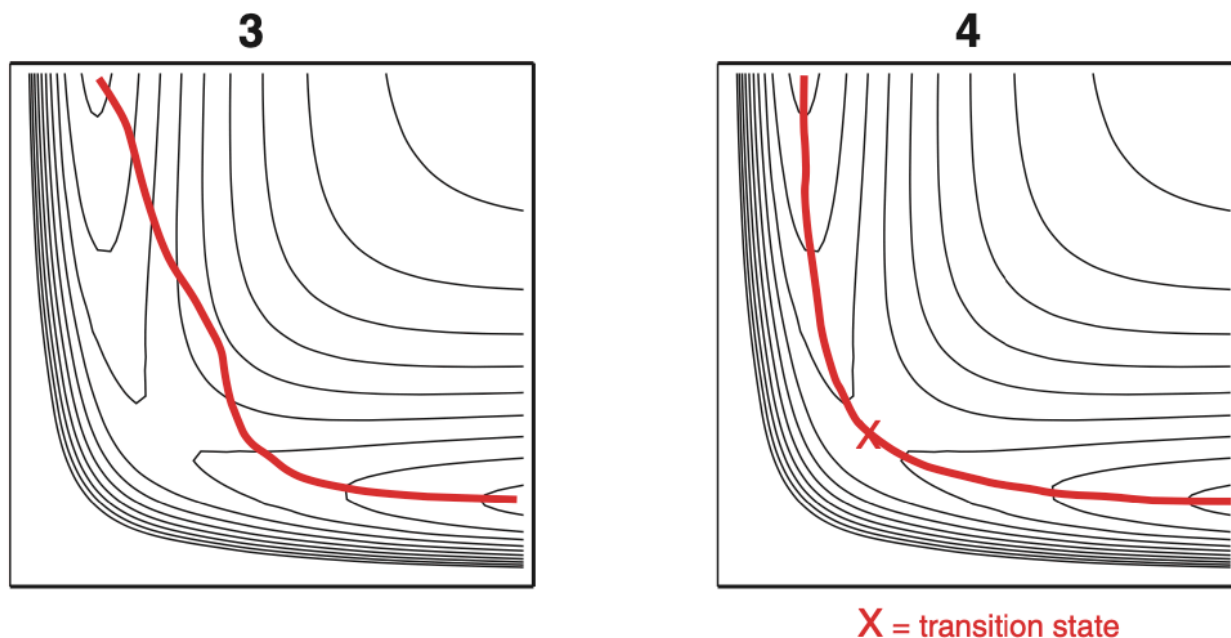
- Route 3 skirts up the left-hand wall of the reactant valley, crosses into the product valley above the pass and then drops down the middle of the product valley. Each route involves a different expenditure of energy, and a different set of movements of the three atoms.



X = transition state

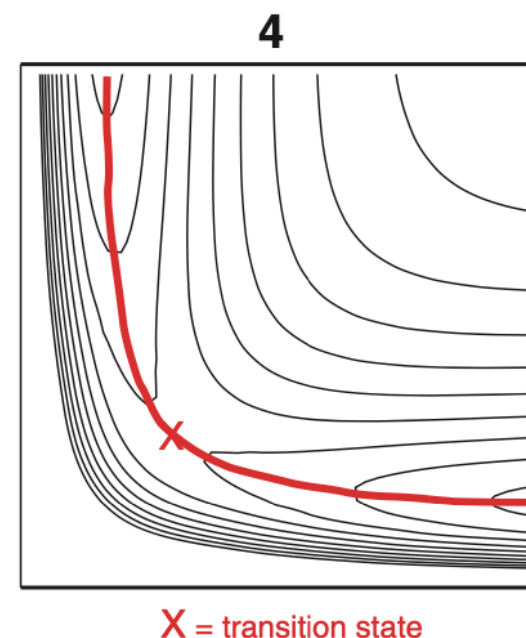
Pathways on the PE surface; the transition state

- Route 4 is a special one. It involves going up the floor of the reactant valley, crossing over the pass at its lowest point and then exiting along the floor the product valley. *It is the pathway that involves the least expenditure of energy and, we shall see, this is the pathway which the reaction takes. The point of highest energy on this pathway is called the transition state;* it corresponds to a molecule in which the A–B bond is partly made and the B–C bond is partly broken. Just exactly where the transition state is depends on the details of the PE surface.



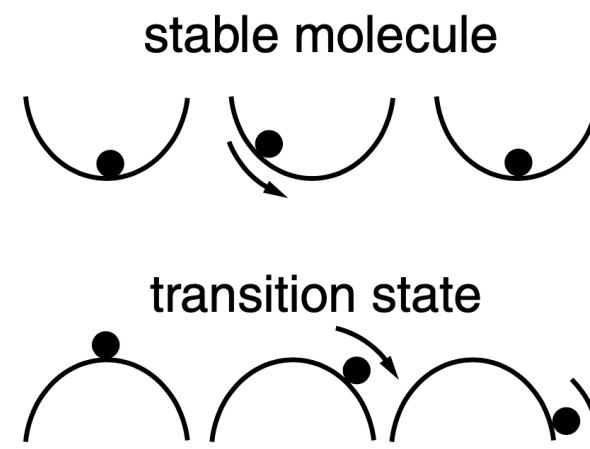
Pathways on the PE surface; the transition state

- This pathway which involves the least expenditure of energy turns out to be the one which the vast majority of successful reactions take; *we will identify this extra energy as the activation energy*. The reason why this pathway is the favoured one is that *for typical values of the activation energy only a tiny fraction of molecular encounters have enough energy to make it to the transition state*. An alternative route, which needs even more energy, will be accessible to even fewer molecules.
- So, for all practical purposes, we can safely assume that *all the reactants which go to products do so via this minimum energy route*.



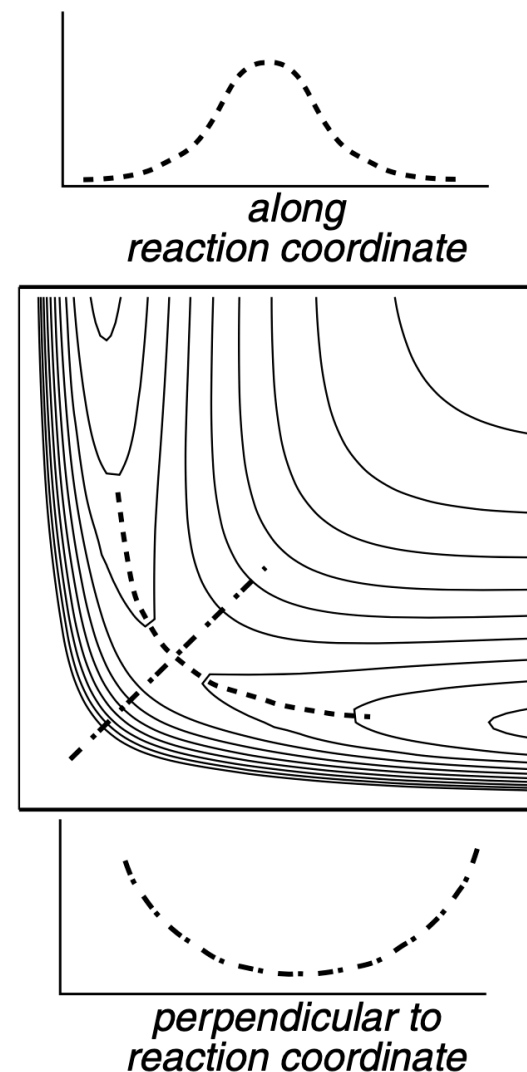
Pathways on the PE surface; the transition state

- *The transition state is not a molecule in the conventional sense.* Molecules exist in potential energy minima; this means then when they distort there is a force which pushes them back to their equilibrium geometry.
- *The PE surface at the transition state is a minima in some directions, but along the path that leads from one valley to the next it is at a maximum.* This means that when the molecule distorts along this path there is no restoring force and the distortion continues; the transition state then ‘falls apart’ either to products or back to reactants.



Pathways on the PE surface; the transition state

- *Along the reaction coordinate the TS exists at a PE maximum*; in other directions the TS is at a minimum, just like other molecules.
- *As it exists at a PE maximum, the transition state only has a fleeting existence*, perhaps of the order of a period of a molecular vibration, $\sim 10^{-13}$ s. The recent advent of lasers capable of producing femto-second pulses has made it possible – just – to study something as short-lived as a transition state.

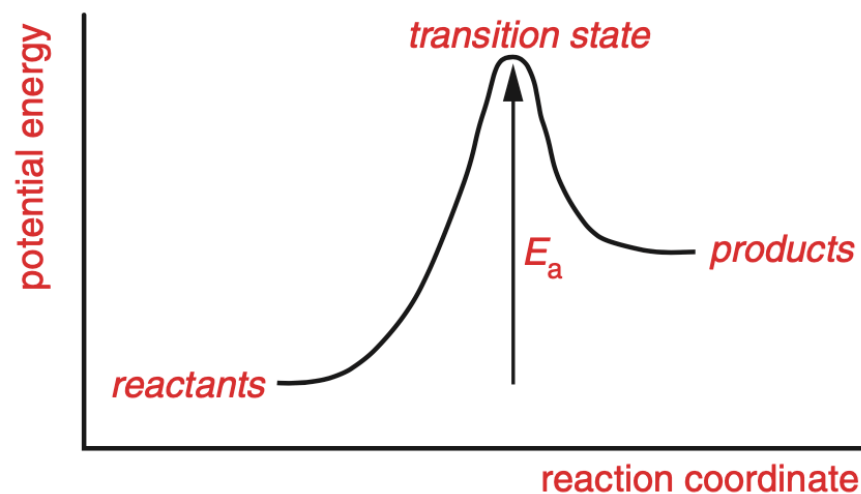


Pathways on the PE surface; the transition state

- The calculations which are needed to map out the PES are time consuming, even for the most powerful computers. A number of surfaces have been mapped out in great detail, including those for the reaction between $\text{H} + \text{H}_2$, $\text{F} + \text{H}_2$ and $\text{H}_2 + \text{OH} \rightarrow \text{H}_2\text{O} + \text{H}$.
- *It is essential to distinguish the transition state from an intermediate.* An intermediate is a molecule like any other, existing in a PE minimum, and detectable by conventional means; it may be highly energetic compared to reactants and products, but it is not unstable in the sense that a transition state is.

The reaction coordinate and the reaction profile

- *The minimum energy pathway from reactants to products is called the reaction co-ordinate.* Of course, travelling along this pathway does not involve simply changing one of the coordinates r_{AB} and r_{BC} ; it involves a complex variation in both. For more complex reactions, the 'coordinate' is a composite of motion in many directions.
- We can plot the energy against the reaction coordinate to give the familiar reaction profile.



- *The transition state exists at the potential energy maxima, and the difference in energy between this maxima and the reactants is called the activation energy, E_a .*